Do students' perceptions of their weight reflect their actual BMI percentile?

12,853
STUDENTS ANALYSED
14,041 raw − 1,188 cleaned

GROUPS · EXPLANATORY — categorical · 5 levels

PerceptionOfWeight

1	Very underweight	n 207
2	Slightly underweight	n 1,436
3	About the right weight	n 7,351
4	Slightly overweight	n 3,318
5	Very overweight	n 541

RESPONSE VARIABLE

BMIPCT BMI percentile

Quantitative measure, ranges 0–100. Higher = heavier relative to peers.

METHOD

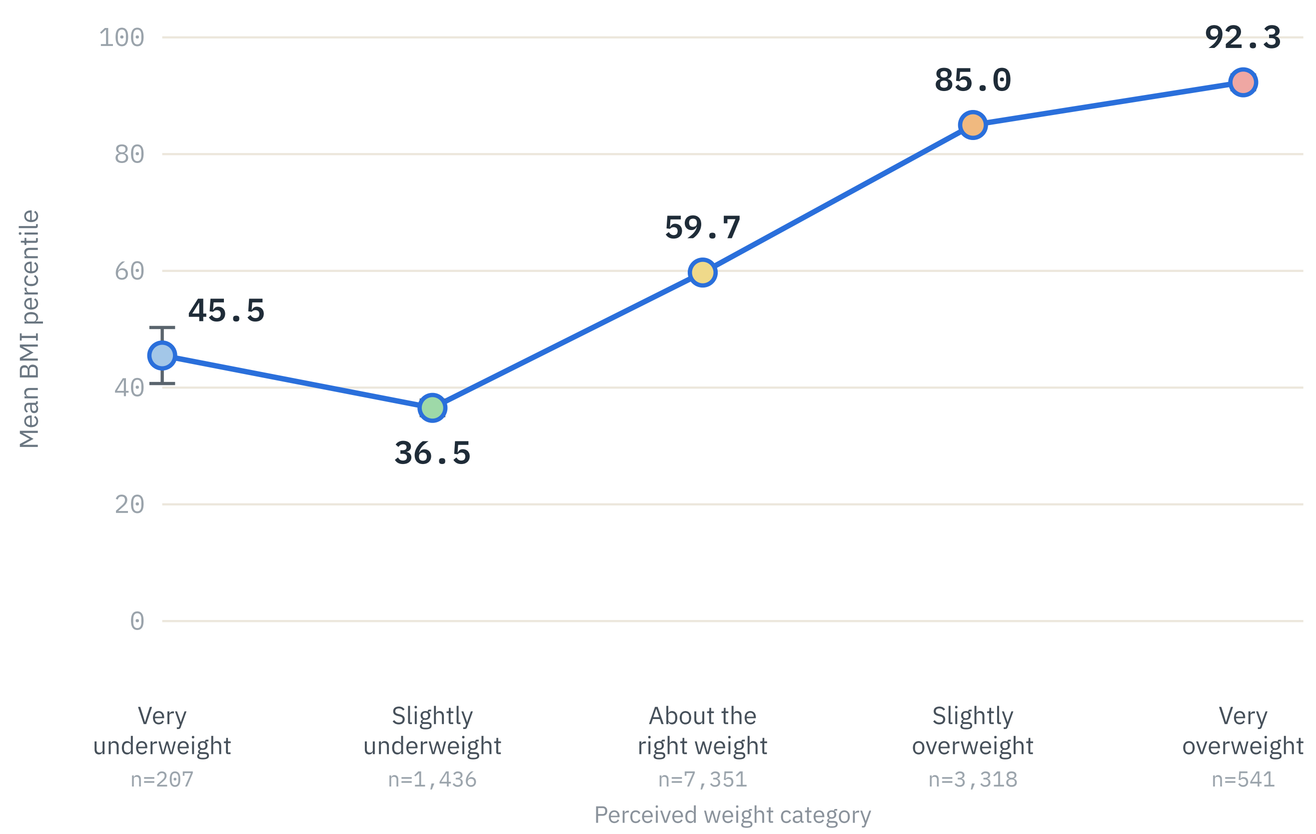
One-way ANOVA

Compares mean BMI percentile across the five perceived-weight groups, then **Tukey HSD** post-hoc for every pair.

H_0: means equal · $\alpha = .05$

Mean BMI percentile is generally higher as perceived weight increases

95% CI error bars



KEY RESULT · ANOVA

$F = 1{,}557$

$p < 0.0001$ → Reject H_0

10/10 TUKEY PAIRS

Every pairwise comparison is significant ($p < .0001$). Largest gap: Very overweight sits **+55.8 pct** above Slightly underweight.

CONCLUSION

Students' perceived weight is clearly **associated** with their actual BMI percentile — those who feel heavier do have higher percentiles. Because YRBS is observational survey data, this is association, not causation.